

Differentiation from State-of-the-Art and Expected Outputs

Differentiation against the closest published systems

Graph type	Bipartite (domain-IP)	Heterogeneous (4 types)		Heterogeneous (4 types)
Update mode	Static (batch)	Static (batch)		Streaming + continuous
Adversarial evaluation	Not reported	Not reported		Multitenancy + adversarial
FPR-anchored evaluation	Yes (0.1% reported)	Not reported		Yes -- five-metric protocol
Operator-validated ground truth	Single deployment	Single campus		Production regular maintenance

Expected Deliverables

**Streaming heterogeneous GNN**
Trained model with continual-learning head, integrated against the live PowerDNS RPZ pipeline.

**Adversarial evaluation harness**
Reproducible attack and defense ablation, packaged as a community resource.

**FPR-anchored evaluation report**
Full five-metric protocol on production telemetry -- directly addressing Survey Section 12.4.

**Peer-reviewed publication**
Indexed journal submission presenting the architecture, robustness study, and operational results.

EVALUATION PROTOCOL

Five Mandatory Metrics — Replacing the Accuracy Fallacy

Survey Section 3.3 and Section 10.2 establish that classification accuracy is operationally meaningless when 99.9% of traffic is benign. This protocol — derived from Survey Section 1.2.4 — is reported for every system in every result table.

O 1 FPR at Threshold $\leq 0.1\%$ Operating-point false-positive rate, anchored to the MANTIS production floor.	O 2 Temporal F1 $T \rightarrow T+1$ F1-score with month T training and month T+1 evaluation. Random splits prohibited.	O 3 AUC-ROC <i>Full range</i> Discrimination across the full operating range, not just at one threshold.	O 4 Inference Latency <i>ms / query</i> Measured on production hardware in the deployment mode, not on benchmark hardware.	O 5 Robustness Δ <i>Attack vs. clean F1</i> Performance degradation under multi-instance attack and recovery with GINGuard.
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Adversarial Threat Model and Integrated Defense

Adversarial robustness is identified as the most consistently neglected evaluation dimension in our published survey (Section 10.3, Section 11). This proposal treats it as a first-class concern.

THE ATTACK • Multi-Instance Structural Perturbation

Nazzei et al., IEEE S&P 2024

Threat model: Adversary controls multiple coordinated domains. Simultaneously manipulates node features and injects fake resolution edges between attacker-controlled and legitimate nodes.

Why GNNs are vulnerable: GNNs assume homophily — connected nodes share labels. Domain registration and DNS query generation are both inexpensive, so this assumption can be violated at low cost.

Reproducibility: Reference implementation publicly available; degrades undefended GNN precision systematically.

THE DEFENSE • GNNGuard Alignment Layer

Zhang and Zitnik, NeurIPS 2020 — model-agnostic, plug-in robustness layer.

Mechanism: Quantifies alignment between graph structure and node-feature similarity. Up-weights edges between similar nodes; prunes edges between dissimilar ones.

Reported recovery: Restores near-clean performance under structural attack across five GNN architectures and four datasets — 15.3% average improvement over prior defenses (NeurIPS 2020 results).

Evaluation plan: Three configurations measured — undefended baseline, GNNGuard-defended, and clean (no-attack) reference — to quantify both attack damage and defense recovery.

You have just completed your project and have to confront the final activities your company requires when putting a project to bed. Which of the following activities is not expected to be part of the project closeout?

Search



25°C
Cloudy

Week 5

1 Point

08:03 PM
14-Sep-26

TEMPORAL DESIGN

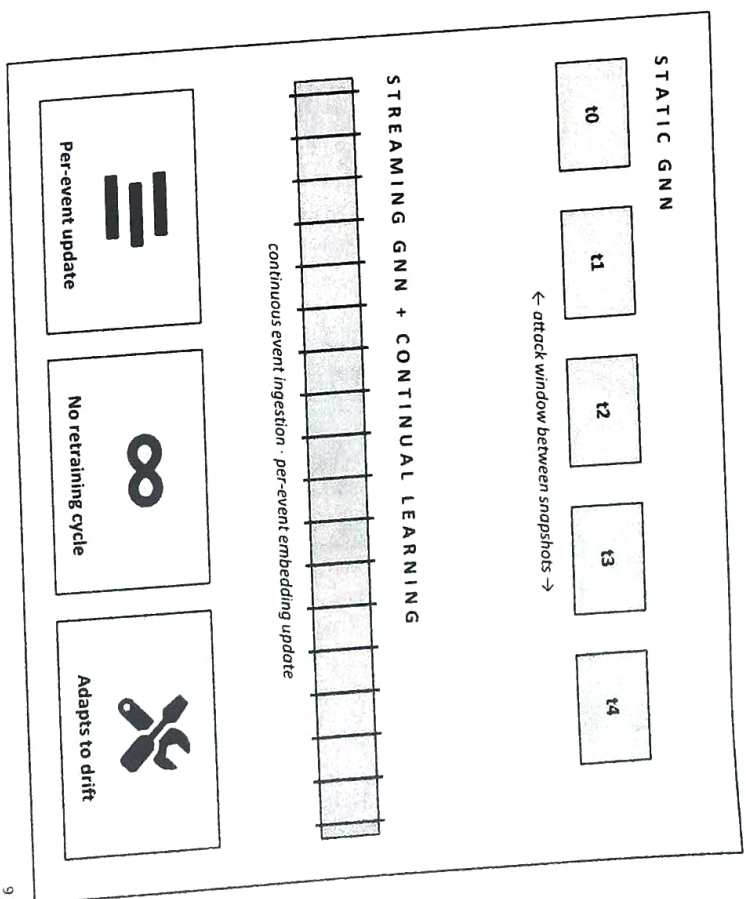
Streaming Updates with Continual Learning

THE PROBLEM WITH STATIC GRAPHS

Fast-Flux adversaries rotate IP associations on TTL timescales of 60 to 300 seconds. Static, batch-mode GNNs see only periodic snapshots and cannot keep up. Worse, the gap between graph rebuilds opens a window for adversarial edge injection that the model cannot detect because the poisoned topology is already baked into the next snapshot.

THE STREAMING APPROACH

A streaming GNN consumes resolution events as they arrive, applies continual-learning objectives (TrafficStream-style) to incorporate new behavior without catastrophically forgetting older patterns, and continuously re-validates topology against learned baselines — closing the temporal gap.



Heterogeneous Graph Design

Lexical models examine each domain in isolation. But attackers reuse infrastructure: hosting, autonomous systems, and TLS certificates persist across lexically unrelated malicious domains. This reuse leaves topological signatures that survive individual takedowns. Heterogeneous graphs formalize this 'guilt-by-association' reasoning, learnable through message passing.

INTELLECTUAL ANCESTRY

MANTIS (S&P 2025) — bipartite domain-IP graph, 0.1% FPR.
DeepDom — heterogeneous information network with SHetGCN.
AGCN-Domain — attention-based hub down-weighting.
Dom-BERT — fusion of graph and language-model embeddings.

PhD Proposal · Ajay Kumar Sharma

HETEROGENEOUS INFORMATION NETWORK



DOMAINS

Queried names; lexical features as node attrs.



IPs

Resolution targets; reputation, hosting density.



AUT. SYSTEMS

BGP-prefix membership; ASN-level enrichment.



TLS CERTS

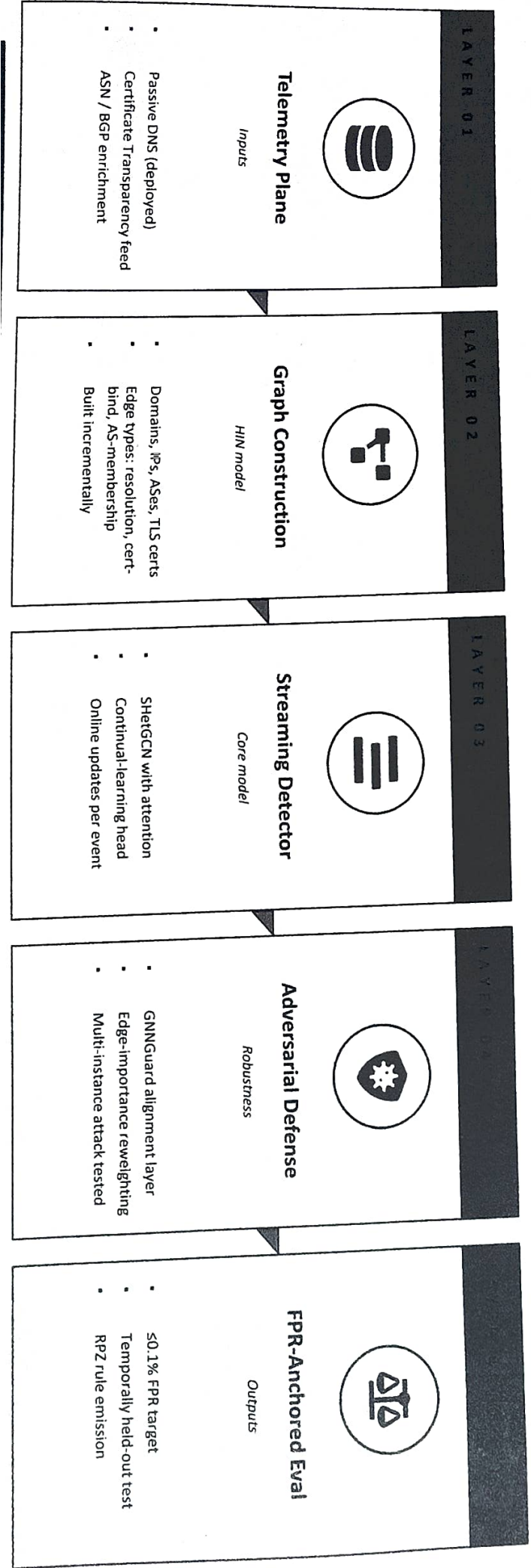
Certificate Transparency log subjects and SANS.

Edge types:

resolution (domain → IP) cert-binding (domain → cert) AS-membership (IP → AS)

PROPOSED FRAMEWORK

Five-Layer Architecture — How the System Fits Together



What Has Been Built — and Why That De-Risks This PhD

Three deliverables already exist. The PhD does not start from a blank slate; it extends a deployed and partially published stock.



Springer LNCS · ICAIS 2026
ACCEPTED

Survey on Malicious Domain Detection

Five-paradigm review of 2021–2026 literature: Transformers, GNNs, LLMs and Foundation Models, encrypted DNS analysis, federated learning. Establishes the gap analysis that this PhD addresses.



Conference paper · Under review

Hybrid DNS Filtering Framework

Multi-stage pipeline: CTI filtering → 1D CNN DGA detection (98.58% accuracy, 0.008 ms inference) → DT/KNN tunneling detection → flow anomaly analysis → SIEM feedback. The lexical and CTI baseline this PhD will compare against.



Production Resolver Deployment

PowerDNS Recursor with Lua postresolve hooks shipping every query to Elasticsearch; RPZ enforcement applied to live traffic across 3,000+ endpoints. The testbed and ground-truth source for this PhD's evaluation.



THESIS STATEMENT

Production-scale DNS threat detection requires moving from lexical, single-paradigm classifiers to streaming, adversarially-robust, heterogeneous graph models that fuse passive DNS with Certificate Transparency, evaluated under FPR-anchored operational metrics rather than accuracy.

This thesis builds, hardens, and evaluates such a system on real production telemetry — directly answering the comparative gaps and future directions identified in our published survey.



Can a streaming, heterogeneous graph model fuse passive DNS with Certificate Transparency to detect threats under FPR-anchored operational metrics?

Detection at low FPR

Adversarial robustness

Predictable graph topology

Survey-Identified Gaps This PhD Will Close

Each gap below is drawn directly from the comparative analysis (Survey Section9–Section10) and future-direction (Survey Section12).

Static graphs cannot track Fast-Flux infrastructure rotation on 60–300 s TTLS (Survey Section5.2, Section12.1).	Streaming heterogeneous GNN with continual-learning head — updates on each resolution event without catastrophic forgetting.
Single-modality lexical detectors miss infrastructure reuse signals (Survey Section5).	Multi-modal fusion of passive DNS, Certificate Transparency logs, AS-level routing, and TLS-cert co-issuance.
Adversarial robustness routinely omitted from evaluation (Survey Section10.3, Section11).	Reproducible multi-instance attack reproduction with GNNGuard alignment defense, ablated independently and combined.
No system reports the five operational metrics together (Survey Section10.2, Section10.4, Section12.4).	FPR-anchored evaluation: FPR $\leq 0.1\%$ target, temporally held-out F1, AUC-ROC, inference latency, adversarial robustness.
Benchmark datasets fragmented; reproducibility weak (Survey Section10.1).	Operator-validated production telemetry from a real recursive resolver deployment — released as a reproducible protocol.

Three Findings That Shape This Proposal

Ajay Sharma, Naveen Kumar, Dharmendra Singh, Rohit Tyagi — Springer LNCS, ICAIS 2026 (accepted & presented): a survey of 2021–2026 work across five paradigms — Transformers, GNNs, LLMs/Foundation Models, encrypted DNS analysis, and federated learning.

01 FINDING

The Accuracy Fallacy is pervasive

In production resolvers, benign queries are >99.9% of traffic. A classifier that labels everything benign scores >99.9% accuracy while detecting nothing. Yet most published systems report only accuracy on class-balanced lab datasets — making operational deployability claims unverifiable.

02 FINDING

Adversarial robustness is treated as optional

Multi-instance structural attacks (Nazal et al., S&P 2024), STING and Graph Backdoor attacks, and federated poisoning are all reproducible. Yet the standard evaluation pipeline ignores them — leaving deployed detectors with documented, exploitable failure modes.

03

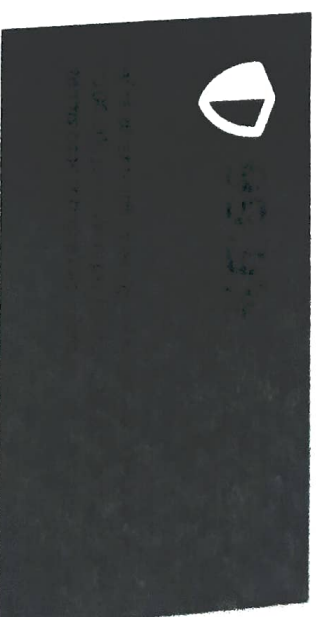
No paper meets the operational bar

No surveyed system simultaneously reports FPR at threshold, F1 on a temporally held-out test set, AUC-ROC, inference latency, and adversarial robustness. The closest operational result — MANTIS at 0.1% FPR — does not evaluate adversarial robustness.

MOTIVATION

Why DNS Threat Detection Demands a Rethink

DNS was designed as a naming infrastructure, not a security boundary. The same resolution machinery that serves legitimate traffic resolves throwaway domains generated by malware. Adversaries exploit this asymmetry through Domain Generation Algorithms, fast-flux infrastructure rotation, encrypted DNS tunneling, and command-and-control beaconing — all of which evade static blocklists and per-query lexical classifiers operating in isolation.



Wordlist DGAs evade entropy

Suppobox, Matsnu and Kraken concatenate dictionary words — defeating entropy-based and N-gram detectors that worked on pseudo-random families.



Encrypted DNS hides payload

DoH and DoT wrap queries inside TLS, removing the payload visibility on which legacy DPI monitors depended.



Adversarial evasion proven

Multi-instance attacks (S&P 2024), STING backdoor and Graph Backdoor have been demonstrated against deployed ML detectors.

